

Jenkins-Practical

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1. CI/CD Pipeline Setup Using Jenkins, Maven, Docker and SCM

- Launch one Ubuntu EC2 instance to set up and configure Jenkins.

The screenshot displays the AWS Management Console interface for an EC2 instance. The top navigation bar shows the AWS logo, a search bar, and the user's account information (PoojaPrakash (7871-6932-0097)). The breadcrumb trail indicates the path: EC2 > Instances > i-0eee779f283178b9c. The main content area is titled "Instance summary for i-0eee779f283178b9c (My Web Server)" and includes an "Info" link. Below the title are buttons for "Connect", "Instance state" (with a dropdown arrow), and "Actions" (with a dropdown arrow). The instance was updated less than a minute ago. The summary is organized into three columns:

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0eee779f283178b9c	13.61.182.133 open address	10.0.7.4
IPv6 address	Instance state	Public DNS
-	Running	ec2-13-61-182-133.eu-north-1.compute.amazonaws.com open address
Hostname type	Private IP DNS name (IPv4 only)	Elastic IP addresses
IP name: ip-10-0-7-4.eu-north-1.compute.internal	ip-10-0-7-4.eu-north-1.compute.internal	It is taking a bit longer than usual to fetch your data
Answer private resource DNS name	Instance type	AWS Compute Optimizer finding
-	t3.micro	Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address	VPC ID	Auto Scaling Group name
It is taking a bit longer than usual to fetch your data	vpc-024f8931266570790 (MyVPC)	-
IAM role	Subnet ID	
-	subnet-02c17cda1d166a85b (Public-Subnet)	

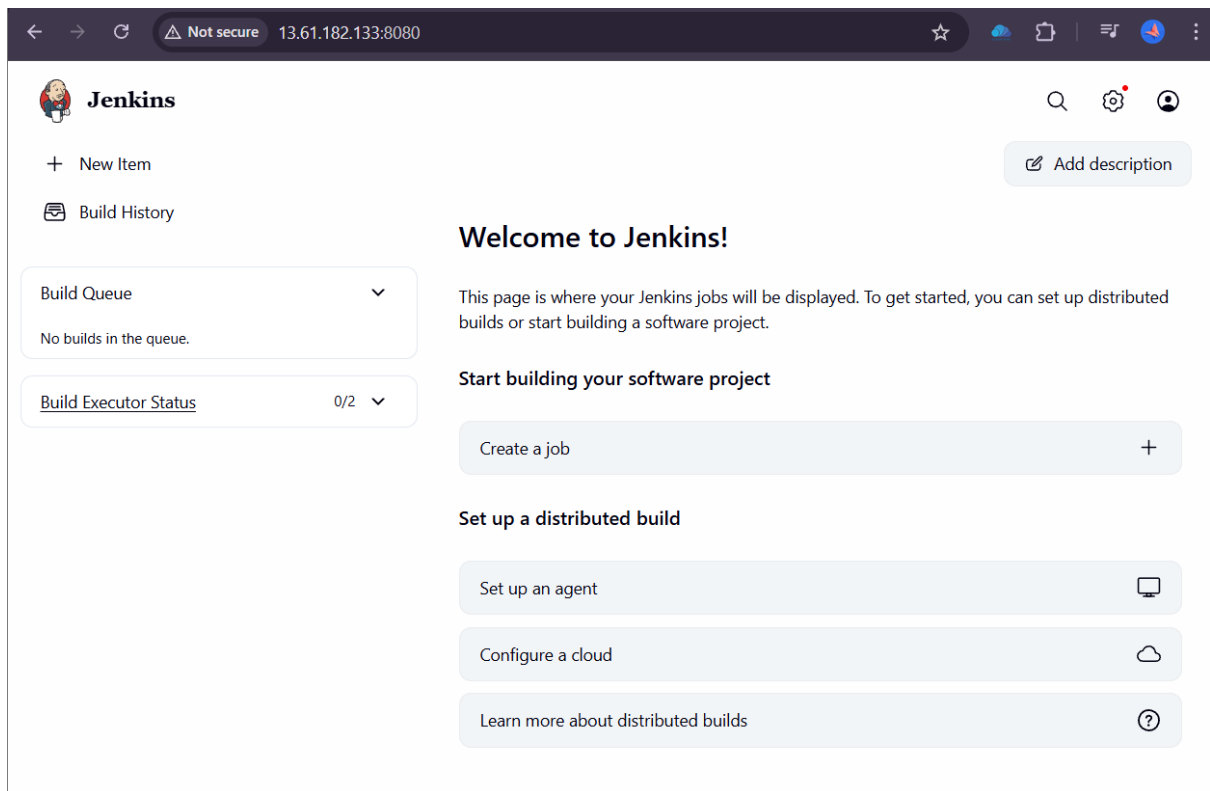
- Connect to the instance using SSH and install the necessary tools like java, maven, git and docker.
- Add the Jenkins user to the Docker group to allow Jenkins to run Docker commands

```
root@ip-10-0-7-4:/home/ubuntu# java -version
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing)
root@ip-10-0-7-4:/home/ubuntu# mvn -version
Apache Maven 3.8.7
Maven home: /usr/share/maven
Java version: 21.0.10, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.14.0-1018-aws", arch: "amd64", family: "unix"
root@ip-10-0-7-4:/home/ubuntu# docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
root@ip-10-0-7-4:/home/ubuntu#
```

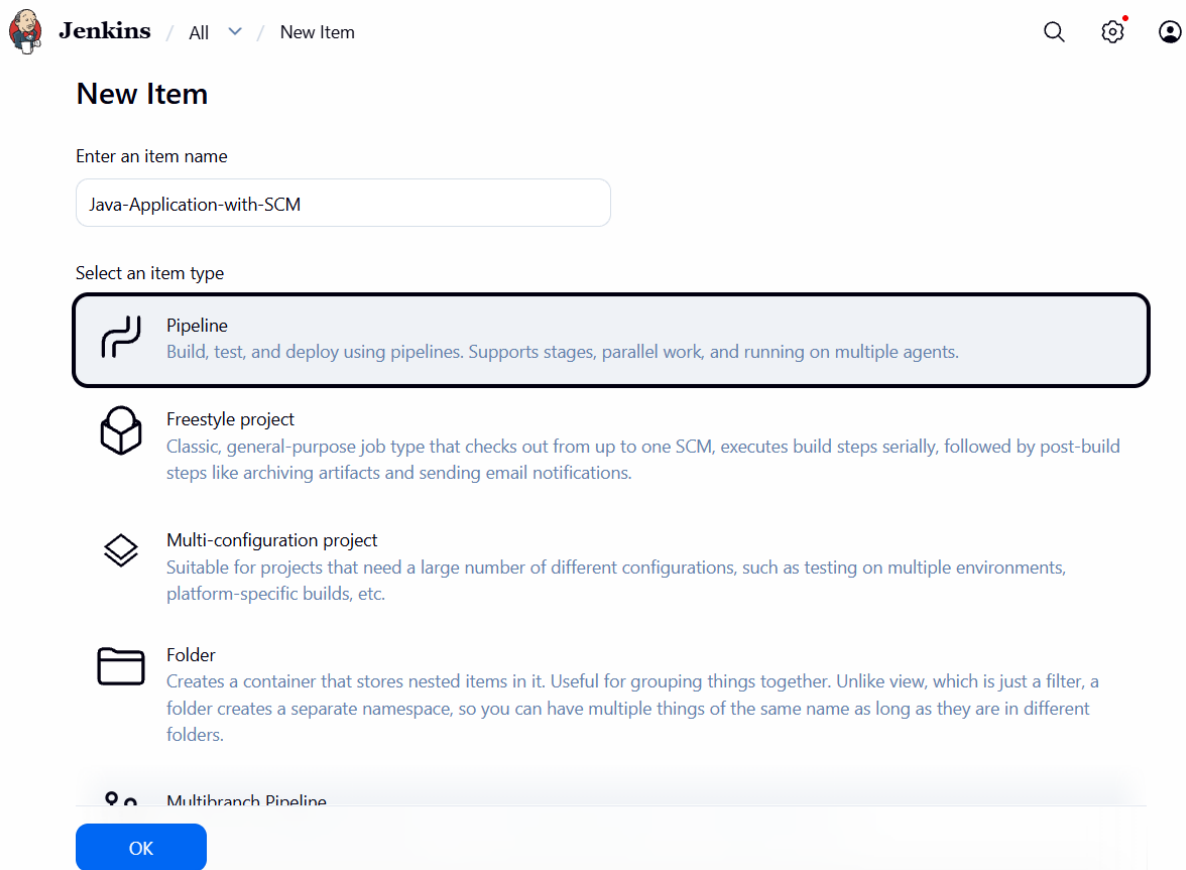
- This application will be built, tested, and containerized using Jenkins.

```
jenkins-tutorial / 3-java-jenkins-docker-app /   
  
PoojaPrakash08 Remove unnecessary blank line in Dockerfile  
  
This branch is 1 commit ahead of theshubhamgour/jenkins-tutorial:main .  
  
NameLast commit message  
..  
srcDirect-Struct mod  
targetDirect-Struct mod  
DockerfileRemove unnecessary blank line in Dockerfile  
JenkinsfileDirect-Struct mod  
README.mdUpdate README.md  
docker-compose.ymlDirect-Struct mod  
pipeline-jenkins-demoUpdate pipeline-jenkins-demo  
pom.xmlDirect-Struct mod
```

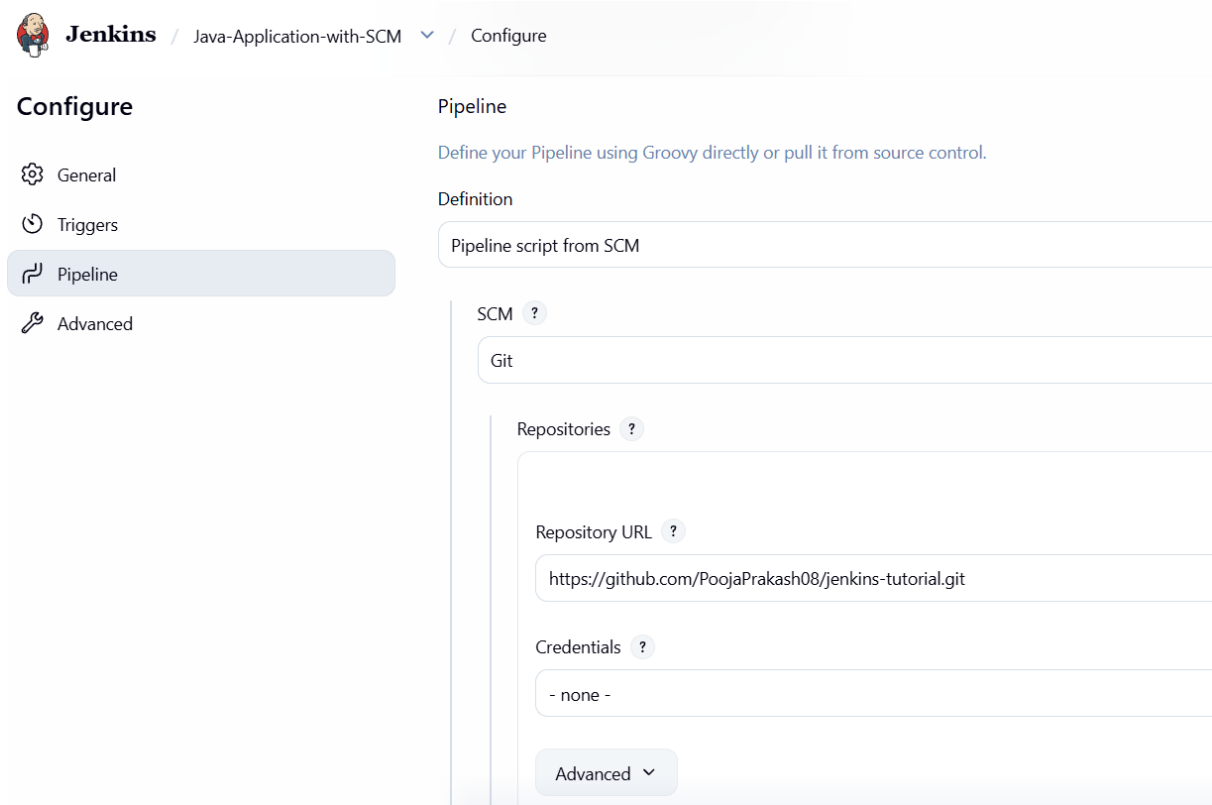
- Install Jenkins on the EC2 server and start the Jenkins service with 8080 port.



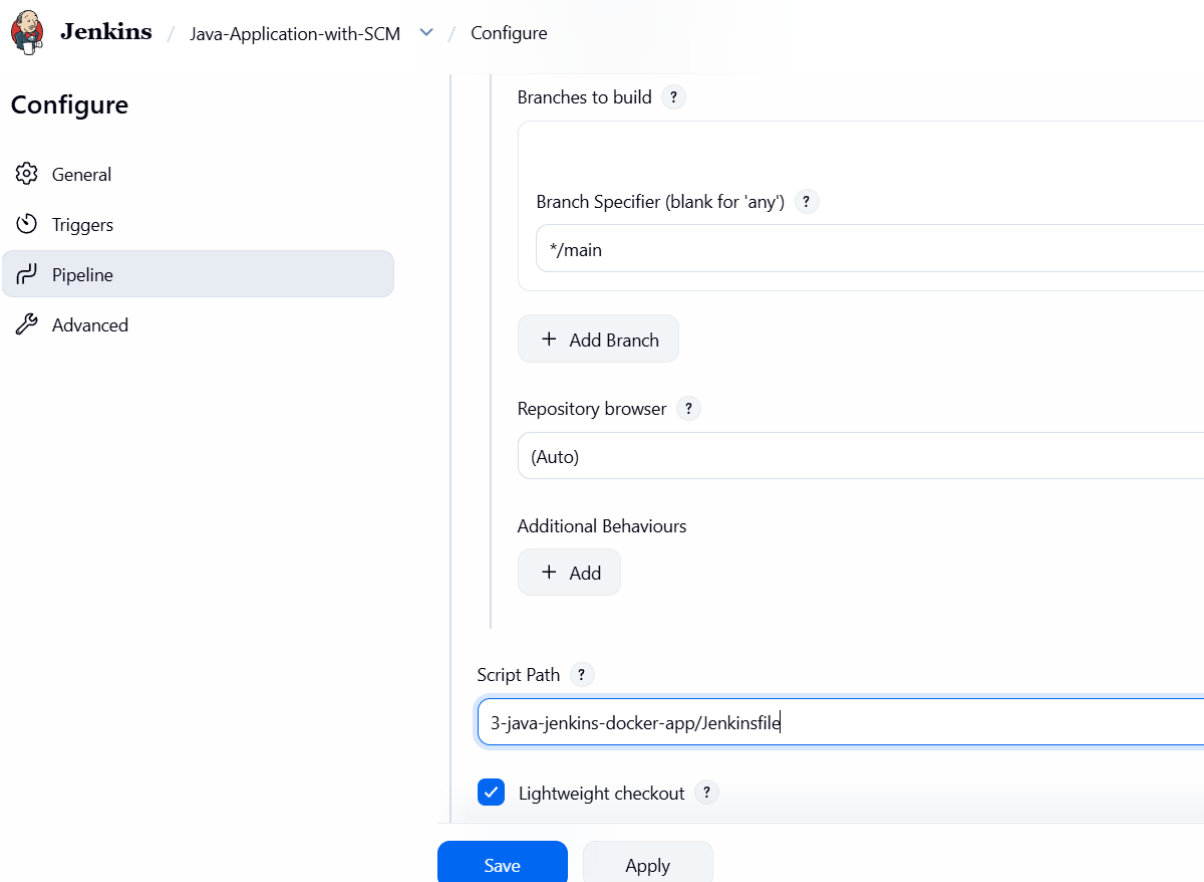
- Create a pipeline job with *java-Application-with-SCM* name



- Select *Pipeline script from SCM*, choose *Git*, enter the repository URL, branch (*main*), provide the Jenkinsfile path, and save



This screenshot shows the Jenkins configuration page for a pipeline. The left sidebar has a 'Configure' header and four options: General, Triggers, Pipeline (which is selected and highlighted), and Advanced. The main area is titled 'Pipeline' and contains the instruction 'Define your Pipeline using Groovy directly or pull it from source control.' Below this, the 'Definition' section has a dropdown menu set to 'Pipeline script from SCM'. The 'SCM' section is set to 'Git'. The 'Repositories' section contains a 'Repository URL' field with the value 'https://github.com/PoojaPrakash08/jenkins-tutorial.git' and a 'Credentials' dropdown set to '- none -'. An 'Advanced' button is visible at the bottom of the configuration area.



This screenshot shows the continuation of the Jenkins pipeline configuration. The left sidebar remains the same. The 'Branches to build' section is expanded, showing a 'Branch Specifier (blank for 'any')' field with the value '*/main'. Below this is a '+ Add Branch' button. The 'Repository browser' is set to '(Auto)'. The 'Additional Behaviours' section has a '+ Add' button. The 'Script Path' field is set to '3-java-jenkins-docker-app/Jenkinsfile'. At the bottom, the 'Lightweight checkout' checkbox is checked. 'Save' and 'Apply' buttons are located at the very bottom of the page.

- The Jenkinsfile builds the application using Maven, runs tests, logs in to Docker Hub using Jenkins credentials, builds the Docker image and runs the container.

```
Code Blame
```

**Raw**

```
1 pipeline {
2     agent any
3     stages {
4         stage('Build') {
5             steps {
6                 dir('3-java-jenkins-docker-app') {
7                     sh 'mvn clean package'
8                 }
9             }
10        }
11        stage('Test') {
12            steps {
13                dir('3-java-jenkins-docker-app') {
14                    sh 'mvn test'
15                }
16            }
17        }
18        stage('Docker Build') {
19            steps {
20                dir('3-java-jenkins-docker-app') {
21                    sh 'docker build -t java-jenkins-docker:latest .'
22                }
23            }
24        }
25        stage('Docker Run') {
26            steps {
27                dir('3-java-jenkins-docker-app') {
28                    sh 'docker run --rm java-jenkins-docker:latest'
29                }
30            }
31        }
32    }
33 }
```


[Jenkins](#) / [Java-Application-with-SCM](#)

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📋 Stages

✎ Rename

❓ Pipeline Syntax

Builds >

Today


#4 4:08 PM

Java-Application-with-SCM

Stage View

Average stage times:
(full run time: ~28s)

#4

Feb 18 21:38

No Changes

Declarative: Checkout SCM	Build	Test	Docker Build	Docker Run
583ms	15s	4s	944ms	1s

Permalinks